A CONNECTED COUNTEREXAMPLE $K_{3,3}$ TO THE AIM BALANCED-CIRCULANT CONJECTURE AND THE CORRECTED CLASSIFICATION

ARGUS OPEN RESEARCH PROJECT

Abstract. Section K.1 of the AIM perfect-graph problem list conjectures that every nonempty balanced circulant graph is isomorphic to $G_B(k,m) = C_{4m}[\overline{K_k}]$ on $4km$ vertices. We verify that the connected circulant $K_{3,3} = \mathrm{Circ}(6;\{1,3\})$ on 6 vertices is balanced (all 9 induced cycles have length 4) yet cannot have $4km$ vertices, and confirm by exhaustive enumeration of all 196,573 nonempty circulants of order $n \le 32$ that the 106 balanced circulants agree exactly with the Morris–Spiga–Webb classification ($G_B(k,m)$ with $m \ge 2$ or $K_{t,t}$ with $t \ge 1$ on each connected component).

1. Main Proposition

Theorem 1.1. *The circulant graph $G = \mathrm{Circ}(6;\{1,3\})$ on $\mathbb{Z}_6$ satisfies:*

(1) *G is connected, 3-regular, vertex-transitive, and bipartite with parts $\{0,2,4\}$ and $\{1,3,5\}$, hence isomorphic to $K_{3,3}$.*
(2) *Every induced cycle of G has length $4 \equiv 0 \pmod 4$ (with zero induced C_6), so G is balanced.*
(3) *Since $|V(G)| = 6$ is not divisible by 4, G is not isomorphic to $G_B(k,m)$ for any positive integers k, m.*

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